

Şevket Kaan Alkır

📍 Ankara, Turkey 🧑 Personal Website ✉ kaanalkir@gmail.com ☎ 0534 679 09 07 in Kaan Alkır

About Me!

Master's student in Mathematics at Bilkent University with a B.Sc. from Ankara University. My research spans inverse reinforcement learning and mean-field games under an average-reward criterion. My paper was accepted to the IEEE CDC 2025, and I am currently working on a manuscript that includes kernel-based methods. I work as a TA supporting undergraduate teaching at the Bilkent University Department of Mathematics.

Education

Bilkent University , <i>M.Sc. in Maths, (Fully Funded)</i>	GPA: 3.09/4.0	2023–Present
◦ In my thesis, I study inverse reinforcement learning for mean-field games under an average-reward criterion. I employ a maximum causal entropy approach to recover expert-consistent policies. Also, model the reward function in a reproducing kernel Hilbert space to capture flexible functional structures.		
Ankara University , <i>B.Sc. in Maths</i>	GPA: 3.44/4.0	2018–2023

Research Interests

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| ◦ Reinforcement Learning. | ◦ Game Theory in Multi-Agent Systems. |
| ◦ Inverse Reinforcement Learning. | ◦ Applications of RL in Complex Systems. |
| ◦ Stochastic Optimization and Control. | ◦ Evolutionary Dynamics in Games. |

Experience

Internship <i>Charles University</i>	<i>Prague, Czech Republic</i> <i>June – Sept. 2022</i>
◦ Used resources "Lectures on Choquet's Theorem, Robert R. Phelps" and "Compact Convex Sets and Boundary Integrals, Erik M. Alfsen"	
Teaching Assistant <i>Bilkent University</i>	<i>Ankara, Turkey</i> <i>Oct. 2023 – Present</i>

Publications and Manuscripts

Inverse Reinforcement Learning for Mean-field Games with Average Reward Criterion <i>Şevket Kaan Alkır*</i> , Naci Saldı. <i>IEEE Conference on Decision and Control</i>	<i>Dec. 2025</i>
Kernel-Based Inverse Reinforcement Learning for Mean-field Games with Average Reward <i>Şevket Kaan Alkır*</i> , Naci Saldı. <i>Manuscript in Preparation</i>	<i>in preparation</i>

Presentations and Talks

[Full list of talks with abstracts and materials](#) [↗](#)

Inverse Reinforcement Learning for Mean-field Games <i>Bilkent University</i>	<i>April 2025</i>
Average-Cost Mean-Field Games: Forward and Inverse Perspectives <i>Özyeğin University</i>	<i>June 2025</i>
Grover's Search Algorithm <i>Bilkent University</i>	<i>Dec. 2024</i>
Frobenius Algebras <i>Bilkent University / Representation Theory Class</i>	<i>Dec. 2023</i>
Multi-Criteria Decision Making: SMART and VIKOR Methods <i>Ankara University</i>	<i>June 2022</i>

Awards and Scholarships

Department of Mathematics Graduate Scholarship
Bilkent University

Sep. 2023 - Present

1001–Scientific and Technological Research Projects Support
Scientific and Technological Research Council of Turkey (TUBITAK)

Oct. 2024 - Present

Projects

Kalman Filter State Estimation

[GitHub](#) 

- Simulated Kalman filter convergence in Python using Riccati updates and visualized estimation errors.

2D Incompressible Fluid Simulation

[Simulation](#) 

- Implemented a numerical solver for the incompressible Navier–Stokes equations using semi-Lagrangian advection and pressure projection on a uniform grid.

Multi-Criteria Decision Making

[Document](#) 

- Applied SMART and VIKOR methods to illustrative case studies, demonstrating practical decision-making under multiple criteria.

Workshops and Events

Probability and Applied Analysis Summer School
Boğaziçi University

June 30 - July 12, 2025

Stochastic Days 2025
Özyeğin University

June 2025

Stochastic Days 2024
Middle-East Technical University

June 2024

Academic References

Naci Saldı, *Bilkent University*
Serdar Yüksel, *Queen's University*

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yuksel@queensu.ca

Skills

Languages: English (IELTS Academic Band 7), Turkish (Native).

Programming Languages: Python, MATLAB, JavaScript. [\[1\]](#)  [\[2\]](#) 